1. **Humanitarian Route & Risk Planning**
   This problem focuses on logistics and physical safety, using the dataset's geographic and casualty features to map out safe zones versus danger zones.

Context / organization	A humanitarian relief agency (e.g., Red Cross, UN OCHA) deploying aid in conflict zones.
Problem statement	Aid agencies struggle to safely deploy medical convoys because violence is localized and unpredictable, putting relief workers at risk.
Stakeholders	Humanitarian logistics managers, field medical teams, and convoy dispatchers.
Business questions	1. Which specific districts (admin2) and coordinates (latitude/longitude) face the most persistent civilian casualties? 2. Do violence hotspots shift predictably following specific trigger events? 3. Which actors (side_a vs. side_b) are most frequently associated with civilian casualties?
Success criteria	The analysis creates a data-backed geographic "risk map" that allows the organization to route convoys away from active, high-casualty hotspots.
Task type: Data mining or Machine Learning or Both	Both. Clustering to identify geographic risk hotspots, and Classification to predict the casualty risk level (low/medium/high) of a region.
Risks and limitations	The dataset may underreport events in totally inaccessible areas (missing population bias), and data only reflects physical conflict events rather than secondary crises like starvation.

2. **Conflict Early Warning System**
   This problem takes an analytical, time-based approach, focusing on trends, timelines, and anomaly detection to predict escalations.

Context / organization	A global peace-monitoring NGO or a data journalism desk.
Problem statement	Peacebuilding organizations lack predictive insights into when small

	skirmishes will escalate into mass-casualty events, delaying urgent civilian warnings.
Stakeholders	Human rights researchers, journalists, and local civilian protection agencies.
Business questions	1. Are there temporal patterns (e.g., days of the week, seasonality) associated with spikes in overall deaths? 2. What sequence of minor events or specific actors typically precede a severe spike in casualties? 3. What anomalies (sudden spikes in violence types) have occurred recently that break historical patterns?
Success criteria	The models successfully identify specific historical precursors to high-casualty events, allowing for a statistical early-warning trigger.
Task type: Data mining or Machine Learning or Both	Data mining. Anomaly detection for unusual spikes, and association rules to link specific actor behaviors to mass casualty outcomes.
Risks and limitations	Media coverage bias might inflate the number_of_sources for specific events while ignoring rural skirmishes, and the dataset is heavily skewed toward one type_of_violence.

3. **Medical Resource Allocation Prediction**
   This problem focuses on healthcare and supply chain management, using casualty estimations and variance to prepare field hospitals.

Context / organization	An international emergency medical coalition (e.g., Doctors Without Borders).
Problem statement	Field hospitals frequently face unexpected surges in mass-casualty events, leading to critical shortages of blood, surgical supplies, and personnel.
Stakeholders	Hospital administrators, emergency surgeons, and global supply chain managers.
Business questions	1. What are the historical maximums for civilian deaths (deaths_civilians) versus

	total deaths in a single localized event? 2. Are certain types of violence (type_of_violence) statistically guaranteed to produce higher casualty rates? 3. How wide is the variance between deaths_low and deaths_high, and what does this uncertainty mean for supply stockpiling?
Success criteria	The analysis accurately segments historical events by severity, providing hospital administrators with a baseline numerical threshold for expected trauma influxes.
Task type: Data mining or Machine Learning or Both	Data mining. Regression to estimate the volume of casualties based on the actors/location, and clustering to segment event severity.
Risks and limitations	Estimating medical needs based solely on deaths ignores severe non-fatal injuries, which require vast medical resources but aren't captured in this specific dataset.

4. **Information Asymmetry and Media Reporting Bias**
   This problem focuses on the reliability of data itself, investigating how media coverage and source verification vary depending on where an event happens and who is involved.

Context / organization	A human rights watchdog group (e.g., Amnesty International or Human Rights Watch).
Problem statement	Human rights organizations struggle to deploy ground investigators efficiently because certain conflict zones and event types suffer from reporting blackouts, leading to unequal documentation of potential violations.
Stakeholders	Human rights investigators, policy researchers, and open-source intelligence (OSINT) analysts.
Business questions	1. Does the volume of casualties (deaths_best) strictly correlate with a higher number of reporting sources (number_of_sources)? 2. Are certain regions (admin1, admin2) systematically underreported (averaging

	only 1 source) despite having frequent, severe incidents? 3. Does the involvement of specific actors (side_a, side_b) decrease the likelihood of an event being verified by multiple sources?
Success criteria	The analysis successfully identifies specific regions and combatant pairings that have statistically suspicious reporting gaps, justifying the deployment of physical investigators to those areas.
Task type: Data mining or Machine Learning or Both	Both. Regression to predict the expected number_of_sources based on casualty size and location, and Anomaly Detection to flag high-casualty events that have surprisingly low reporting.clustering to segment event severity.
Risks and limitations	The number_of_sources metric only counts the specific sources validated by the UCDP methodology, which may not represent the entirety of global or local media coverage.

Variables needed for this option: number_of_sources, deaths_best, admin1, admin2, side_a, side_b, and status.

Additional link:
https://conflictstats.com/country/israel-and-palestine/?fbclid=IwY2xjawVBYahleHRuA2FlbQIxMABwZG9mBWJyaWQRMUVDOWFHVXJ4SnJqek9VTk5zcnRjBmFwcF9pZBAyMjIwMzkxNzg4MjAwODkyAAEez-vyx3TyaZ4dDjM0T8I3y_-Z8S5tCaG-hXmvMebQVLeCEbVicelWBG3NJuk_aem_NVte-_0YE6EOHD3eI9kFag